

```
data collected Mon, 09 Jul 2018 16:09:09
```

```
File 'sshd_dsa_key'
```

```
status Accessible
monitoring status Monitored
monitoring mode active
on reboot start
permission 600
uid 0
gid 0
size 227 B
timestamp Thu, 05 Apr 2018 13:18:18
checksum
```

```
abbbf249f7ca23dd6f8186072dc0173a3 (MD5)
```

```
data collected Mon, 09 Jul 2018 16:09:09
```

```
File 'sshd_rc'
```

```
status Accessible
monitoring status Monitored
monitoring mode active
on reboot start
permission 644
uid 0
gid 0
size 3.2 kB
timestamp Thu, 05 Apr 2018 13:18:17
checksum
```

```
bbad7ed242a834e831c7066901cee49e (MD5)
```

```
data collected Mon, 09 Jul 2018 16:09:09
```

```
System 'debian'
```

```
status Running
monitoring status Monitored
monitoring mode active
on reboot start
load average [0.00] [0.00] [0.00]
cpu 0.0%us 0.0%sy 0.0%wa
memory usage 47.6 MB [4.8%]
swap usage 0 B [0.0%]
uptime 1h 6m
boot time Mon, 09 Jul 2018 15:03:06
data collected Mon, 09 Jul 2018 16:09:09
```

```
become: yes
become_method: sudo
gather_facts: true
tags: [nginx]
```

```
tasks:
```

```
- name: Update the software package repository
```

```
apt:
```

```
update_cache: yes
```

```
- name: Install nginx
```

```
package:
```

```
name: "{{ item }}"
```

```
state: latest
```

```
with_items:
```

```
- nginx
```

```
- name: Start nginx
```

```
service:
```

```
name: nginx
```

```
state: started
```

```
- wait_for:
```

```
port: 80
```

```
- name: Add nginx monitoring
```

```
blockinfile:
```

```
path: /etc/monit/monitrc
```

```
marker_begin: "nginx BEGIN"
```

```
marker_end: "nginx END"
```

```
block: |
```

```
check process nginx with pidfile /var/run/nginx.pid
```

```
group www
```

```
group nginx
```

```
start program = "/etc/init.d/nginx start"
```

```
stop program = "/etc/init.d/nginx stop"
```

```
if 5 restarts with 5 cycles then timeout
```

```
depend nginx_bin
```

```
depend nginx_rc
```

```
check file nginx_bin with path /usr/sbin/nginx
```

```
group nginx
```

```
include /etc/monit/templates/rootbin
```

```
check file nginx_rc with path /etc/init.d/nginx
```

Nginx

We can also set up Nginx on the guest VM and monitor it using Monit. The software package repository is first updated and Nginx is then installed. The service is started and we wait for it to listen on port 80. The Nginx monitoring check is then added to the Monit configuration file and the Monit service is restarted. The Ansible playbook for the above tasks is provided below, for reference:

```
- name: Configure ssh monitoring
hosts: debian
```

Monit Service Manager						
Monit is gathering on debian and monitoring:						
System	Status	Load	CPU	Memory	Swap	
debian	Running	[0.00] [0.00] [0.00]	0.0%us, 0.0%sy, 0.0%wa	0.1% [0.0 MB]	0.0% [0.0]	
Process	Status	Uptime	CPU Total	Memory Total		
sshd	Running	1h 30m	0.0%	3.4% [0.7 MB]		
nginx	Running	22m	0.0%	0.0% [0.0 MB]		
File	Status	Size	Permission	UID	GID	
sshd_bin	Accessible	772.5 kB	0755	0	0	
sshd_rcs_bin	Accessible	1.6 kB	0600	0	0	
sshd_dsa_key	Accessible	227 B	0600	0	0	
sshd_rcs	Accessible	0.2 kB	0644	0	0	
nginx_bin	Accessible	1.0 MB	0755	0	0	
nginx_c	Accessible	4.6 kB	0755	0	0	

Figure 3: Monit Web UI with SSH and Nginx